

I Introduction

Snow is one of the major component of the cryosphere. Its global distribution on both ocean and land areas strongly depends on seasonality, latitude, and elevation. It occurs in three forms, mainly: permanent snow on ice sheets and glaciers, snow on sea ice, and seasonal snow cover. The latter is estimated to account for 1.3% (summer) to 30.6% (winter) of the Northern Hemisphere land surface (Vaughan et al., 2013), with the 1967-2026 mean snow cover extent (SCE hereafter) in winter being around $39.79 \times 10^6 \text{ km}^2$ (Robinson & Estilow, 2012). Sea ice is almost completely covered by snow during the winter season; its peak concentration, therefore representative of sea ice specific SCE, occurs in March in the Northern Hemisphere ($15.03 \times 10^6 \text{ km}^2$) and in September in the Southern Hemisphere ($18.61 \times 10^6 \text{ km}^2$) (Fetterer et al., 2025; Warren et al., 1999). Finally, perennial snow mean extent over the 2000-2022 period is estimated at $16.47 \times 10^6 \text{ km}^2$ from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS), including glaciers, ice caps, Greenland and Antarctica ice sheets (Johnston et al., 2024).

These several tens of millions of snow-covered squared kilometres play a major role in freshwater resources availability, ecosystems equilibrium and, particularly, climate regulation. In fact, among its many physicochemical properties, snow represents one of the strongest reflective materials on Earth's surface: its interaction with the incident solar radiation has therefore a relevant impact on our Planet's energy budget (National Snow and Ice Data Center, 2024). The fraction of solar radiation delivered to a body that is diffusely reflected by it is called albedo; for snow, this amount is roughly close to 1, since its highest reflectivity occurs in the range of visible light, *i.e.* where solar radiation peaks (Kokhanovsky & Zege, 2004).

II Material and methods

1 Theoretical backgrounds

!!! This risks to become "textbook"-like - then, must be SUPER ADAPTED TO MY SPECIFIC GOAL AND BE VERY PERSONAL AND NOT GENERAL AT ALL - PASSIONATING ! write it after matmethods and results

2 Material and instruments

2.1 *In situ*: the Multiband albedometer

2.1.a *The stations*

Cap Prud'Homme

EAIIST

2.2 *In situ*: the Solalb albedometer

2.2.a *Campaigns*

2.3 Satellite imagery: Sentinel-3 Ocean Land and Colour Instrument

2.3.a *Generalities*

Orbit, resolution, swath, revisit time etc.

2.3.b *Data availability*

3 The workflow

Nice flowchart

3.0.a

III Results

PRESENTING MY RESULTS CSI Problem horizontality/titlting ? Calibrations analysis/response symmetry

Do not simply list the results, it can be boring If the comclution from a result is very fast, just put it in the results

IV Discussion

CONTEXTING MY RESULTS IN SCIENTIFIC LITERATURE Typically no plots

V Conclusion

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